APPENDIX I

# MATLAB® Functions (not including those listed in the "Explore Other Interesting Features" sections)

**abs** absolute value
**all** true if all elements in the input argument are true
**angle** angle of a complex number
**any** true if any element in the input argument is true
**area** filled two-dimensional area plot
**asin** arcsine in radians
**asind** arcsine in degrees
**asinh** inverse hyperbolic sine in radians
**axis** sets limits on axes for a plot
**bar** two-dimensional bar chart
**bar3** three-dimensional bar chart
**bar3h** three-dimensional horizontal bar chart
**barh** two-dimensional horizontal bar chart
**blanks** creates a string of all blank spaces
**cast** casts a variable to a specified type
**categorical** converts cell array to categorical array
**categories** lists categories from a categorical array
**ceil** rounds toward infinity
**cell** creates a cell array
**celldisp** displays contents of a cell array
**cellplot** displays contents of a cell array in boxes
**cellstr** converts from a character matrix to a cell array of strings
**char** creates a character matrix
**checkcode** displays Code Analyzer results for code files
**class** returns the type or class of the input argument
**clear** clears variable(s) from the workspace
**clf** clears the figure window
**clock** stores the current date and time in a vector
**collect** collects like terms in a symbolic math expression
**colorbar** displays a color scale in a plot

**colormap** returns the current colormap, or sets a matrix to be the current colormap
**comet** animated two-dimensional plot
**comet3** three-dimensional animated plot
**complex** creates a complex number
**conj** complex conjugate
**cross** cross product
**cummax** cumulative, or running, maximum of a vector or columns of a matrix
**cummin** cumulative, or running, minimum of a vector or columns of a matrix
**cumprod** cumulative, or running, product of a vector or columns of a matrix
**cumsum** cumulative, or running, sum of a vector or columns of a matrix
**cylinder** returns three-dimensional data vectors to create a cylinder
**date** stores the current date as a string
**dbcont** continue executing code in debug mode
**dbquit** quit debug mode
**dbstep** step through code in debug mode
**dbstop** set a breakpoint in debug mode
**deblank** gets rid of trailing blanks in a string
**deg2rad** converts from degrees to radians
**demo** shows MATLAB Examples in the Help Browser
**det** finds the determinant of a matrix
**diag** returns the diagonal of a matrix, or creates a diagonal matrix
**diff** finds differences between consecutive elements; used to approximate derivatives
**disp** simple display (output)
**doc** brings up a documentation page
**dot** dot product
**double** converts to the type double
**echo** toggle; displays all statements as they are executed
**end** ends control statements and functions; refers to last element
**error** displays an error message
**eval** evaluates a string as a function or command
**exit** quits MATLAB
**exp** exponential function
**expand** expands a symbolic math expression
**eye** creates an identity matrix
**ezplot** simple plot function that plots a function without the need for data vectors
**factor** factors a symbolic math expression
**factorial** factorial of an integer *n*, is 1*2*3*…**n*
**false** equivalent to logical(0); creates an array of false values
**fclose** closes an open file
**feof** true if the specified file is at the end-of-file

**feval** evaluates a function handle on a string as a function call
**fgetl** low-level input function reads one line from a file as a string
**fgets** same as fgetl but does not remove newline characters
**fieldnames** returns the names of fields in a structure as a cell array of strings
**figure** create or refer to Figure Windows
**find** returns indices of an array for which a logical expression is true
**fix** rounds toward zero
**flip** flips an array, either left to right or up to down
**fliplr** flips columns of a matrix from left to right
**flipud** flips rows of a matrix up to down
**floor** rounds toward negative infinity
**fopen** low-level file function; opens a file for a specified operation
**format** many options for formatting displays
**fplot** plots a function passed as a function handle
**fprintf** formatted display (output); writes either to a file or to the screen (the default)
**fscanf** low-level file input function; reads from a file into a matrix
**func2str** converts from a function handle to a string
**fzero** attempts to find a zero of a function, given the function handle
**gca** handle to the current axes
**gcf** handle to the current figure
**get** gets properties of a plot object
**getframe** gets a movie frame, which is a snapshot of the current plot
**ginput** gets graphical coordinates from a mouse click
**grid** plot toggle; turns grid lines on or off
**groot** handle of graphics root object (screen)
**gtext** allows the user to place a string on a plot in location of a mouse click
**help** displays help information for built-in or user-defined functions, or scripts
**histogram** plot function: plots a histogram
**hold** plot toggle; freezes plot in Figure Window so the next will be superimposed
**i** constant for the square root of negative one
**im2double** converts an image matrix to type double
**imag** imaginary part of a complex number
**image** displays an image matrix
**imread** reads in an image matrix
**imshow** displays an image
**imwrite** writes a matrix in an image format
**inf** constant for infinity
**input** prompts the user and reads user's input
**int** symbolic math integration
**int16** converts a number to a 16-bit signed integer
**int2str** converts from an integer to a string storing the integer

**int32** converts a number to a 32-bit signed integer
**int64** converts a number to a 64-bit signed integer
**int8** converts a number to an 8-bit signed integer
**intersect** set intersection
**intmax** largest value possible in a specified integer type
**intmin** smallest value possible in a specified integer type
**inv** inverse of a matrix
**isa** true if the input argument is the specified class
**isbanded** true if the input matrix is banded
**iscellstr** true if the input argument is a cell array storing only strings
**ischar** true if the input argument is a string, or character vector
**isdiag** true if the input matrix is a diagonal matrix
**isempty** true if the input argument is an empty vector or empty string
**isequal** true if two array arguments are equal element-by-element
**isfield** true if a string is the name of a field within a structure
**iskeyword** true if the string input argument is the name of a keyword
**isletter** true if the input argument is a letter of the alphabet
**ismember** set function receives two sets; true for every member of first set also in second
**isreal** true if input argument is a real number (not complex)
**issorted** true if the input vector is sorted in ascending order
**isspace** true if the input argument is a white space character
**isstrprop** true if the string argument is a specified property
**isstruct** true if the input argument is a structure
**issymmetric** true if the input matrix is symmetric
**istril** true if the input matrix is lower triangular
**istriu** true if the input matrix is upper triangular
**j** constant for the square root of negative one
**jet** returns all or part of the 64 colors in the jet colormap
**legend** displays a legend on a plot
**length** length, or number of elements, in a vector; largest dimension for a matrix
**limit** computes limit of a symbolic math expression
**line** graphics primitive object that creates a line
**linspace** creates a vector of linearly-spaced values
**load** inputs a file into a matrix, or reads variables from a.mat file (the default)
**log** natural logarithm
**log10** base 10 logarithm
**log2** base 2 logarithm
**logical** converts numbers to the type logical
**loglog** plot function that uses logarithmic scales for $x$ and $y$ axes
**logspace** creates a vector of logarithmically spaced values
**lookfor** looks for a string in the H1 comment line in files

**lower** converts letters to lower-case in a string
**max** the maximum value in a vector, or for every column in a matrix
**mean** the mean (average) of values in a vector, or every column in a matrix
**median** the median (middle) value in a sorted vector, or for every column in a matrix
**menu** displays a menu of push buttons and returns number of choice (not recommended)
**mesh** three-dimensional mesh surface plot
**meshgrid** creates $x$ and $y$ vectors to be used in images or as function arguments
**methods** displays methods of a class
**min** the minimum value in a vector, or for every column in a matrix
**mod** modulus after division
**mode** the maximum value in a vector, or for every column in a matrix
**movegui** moves a Figure Window within the screen
**movie** plays a movie, or sequence of screen shots
**namelengthmax** the maximum length of identifier names
**NaN** mathematics constant for "Not a Number"
**nargin** number of input arguments passed to a function
**nargout** number of output arguments expected to be returned by a function
**nthroot** $n$th root of a number
**num2str** converts a real number to a string containing the number
**numden** symbolic math function, separates the numerator and denominator of a fraction
**numel** total number of elements in a vector or matrix
**ones** creates a matrix of all ones
**parula** returns all or part of the 64 colors in the parula colormap
**patch** graphics primitive object that creates a filled-in two-dimensional polygon
**pi** constant for $\pi$
**pie** creates a two-dimensional pie chart
**pie3** creates a three-dimensional pie chart
**pink** returns all or part of the 64 colors in the pink colormap
**plot** simple plot function, plots 2D points; markers, color, etc. can be specified
**plot3** simple three-dimensional plot function, plots three-dimensional points
**plus** the functional form of the addition operator
**polarplot** plot function for complex numbers, plots the magnitude and angle
**poly2sym** converts a vector of coefficients of a polynomial to a symbolic expression
**polyder** derivative of a polynomial
**polyfit** fits a polynomial curve of a specified degree to data points
**polyint** integral of a polynomial
**polyval** evaluates a polynomial at specified value(s)
**pretty** displays a symbolic expression using exponents

**print** prints or saves a figure or image
**prod** the product of all values in a vector, or of every column in a matrix
**profile** toggle; the Profiler generates reports on execution time of code
**properties** displays properties of a class
**quad** integration using Simpson's method
**quit** quits MATLAB
**rad2deg** converts from radians to degrees
**rand** generates uniformly distributed random real number(s) in the open interval (0,1)
**randi** generates random integer(s) in the specified range
**randn** generates normally distributed random real numbers
**real** real part of a complex number
**rectangle** graphics primitive to create a rectangle; curvature can vary
**rem** remainder after division
**repelem** replicates elements in a matrix; creates *m x n* copies of each
**repmat** replicates a matrix; creates *m x n* copies of the matrix
**reshape** changes dimensions of a matrix to any matrix with the same number of elements
**rgb2gray** converts an RGB image matrix to grayscale
**rmfield** removes a field from a structure
**rng** random number generator, sets the seed for random functions and gets the state
**roots** roots of a polynomial equation
**rot90** rotates a matrix 90 degrees counter-clockwise
**round** rounds a real number toward the nearest integer
**rref** puts an augmented matrix in reduced row echelon form
**save** writes a matrix to a file or saves variables to a.mat file
**semilogx** plot function, uses a scale for logarithmic $x$ and a linear scale for $y$
**semilogy** plot function, uses a linear scale for $x$ and a logarithmic scale for $y$
**set** sets properties of a plot object
**setdiff** set function, returns elements that are in one vector, but not in another
**setxor** set exclusive or, returns the elements that are not in the intersection of two sets
**sign** signum, returns −1, 0, or 1
**simplify** simplifies a symbolic math expression
**sin** sine in radians
**sind** sine in degrees
**single** converts a number to the type single
**sinh** hyperbolic sine in radians
**size** returns the dimensions of a matrix
**solve** symbolic math function to solve an equation or simultaneous equations
**sort** sorts the elements of a vector (default is ascending order)
**sortrows** sorts the rows of a matrix; for strings result in an alphabetical sort

**sound** sends a sound signal (vector of amplitudes) to an output device
**sphere** returns three-dimensional data vectors to create a sphere
**spiral** creates a square matrix of integers spiraling from 1 in the middle
**sprintf** creates a formatted string
**sqrt** square root
**std** standard deviation
**stem** two-dimensional stem plot
**stem3** three-dimensional stem plot
**str2double** converts from a string containing a number to a double number
**str2func** converts a string to a function handle
**str2num** converts from a string containing number(s) to a number array
**strcat** horizontal string concatenation
**strcmp** string compare, used instead of equality operator for strings
**strcmpi** string compare, ignoring case
**strfind** find a substring within a longer string
**strjoin** concatenates strings in a cell array into a long string
**strncmp** string compare the first *n* characters of strings
**strncmpi** string compare the first *n* characters, ignoring case
**strrep** replace all occurrences of one substring with another within a longer string
**strsplit** splits a string into elements in a cell array
**strtok** breaks one longer string into two shorter strings, with all characters retained
**strtrim** deletes both leading and trailing blanks from a string
**struct** creates a structure by passing pairs of field names and values
**subplot** creates a matrix of plots in the Figure Window
**subs** substitutes a value into a symbolic math expression
**sum** the sum of the values in a vector or of every column in a matrix
**summary** shows variables and statistics for a table
**surf** three-dimensional surface plot
**sym** creates a symbolic variable or expression
**sym2poly** converts a symbolic expression to a vector of coefficients for a polynomial
**syms** creates multiple symbolic variables
**table** creates a table data structure
**text** graphics primitive object to put a string on a plot
**textscan** file input function, reads from a file into a cell array of column vectors
**tic/toc** used to time code
**timeit** times a function execution
**title** writes a string as a title on a plot
**trace** the trace (sum of values on the diagonal) of a matrix
**trapz** trapezoidal rule to approximate the area under a curve
**tril** converts a matrix to a lower triangular matrix

**triu** converts a matrix to an upper triangular matrix
**true** equivalent to logical(1), creates a matrix of all true values
**type** display the contents of a file in the Command Window
**uibuttongroup** groups together button objects
**uicontrol** basic function to create graphical user interface objects of different styles
**uint16** converts a number to a 16-bit unsigned integer
**uint32** converts a number to a 32-bit unsigned integer
**uint64** converts a number to a 64-bit unsigned integer
**uint8** converts a number to an 8-bit unsigned integer
**uipanel** groups together graphical user interface objects
**union** set function, the union of two sets
**unique** returns all of the unique values within a set (vector)
**upper** converts all letters to upper-case
**var** variance
**varargin** built-in cell array to store input arguments
**varargout** built-in cell array to store output arguments
**who** displays variables in the base workspace
**whos** displays more information on the variables in the base workspace
**xlabel** puts a string as a label on the *x* axis of a plot
**xlsread** reads from a spreadsheet with filename.xls
**xlswrite** writes to a spreadsheet with filename.xls
**xor** exclusive or, true if only one argument is true
**ylabel** puts a string as a label on the *y* axis of a plot
**zeros** creates a matrix of all zero values
**zlabel** puts a string as a label on the *z* axis of a three-dimensional plot